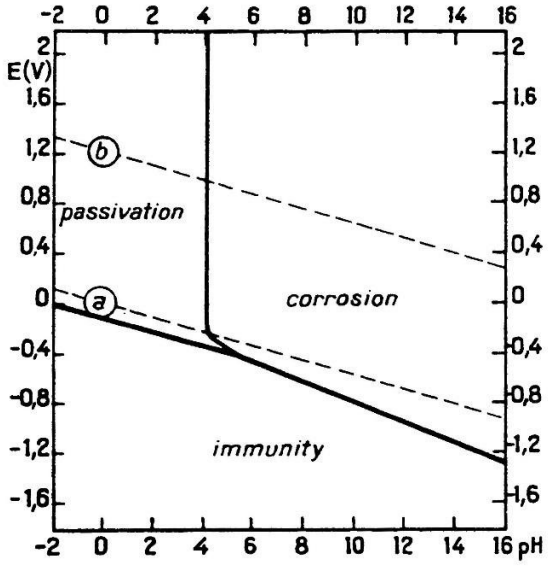
Inorganic Cumulative Exam
April 25 2025

Answer any 5 of the following 10 questions (20 points each). Any answers provided beyond the requested five will not be graded. Remember: Your answers to these questions are meant to provide evidence of broad, graduate-school level chemistry knowledge. Consider this in your choice of terminology and examples as answers to these questions.

1. Describe the water gas shift reaction, including the reactant(s), product(s), and commonly used catalysts for the reaction.
2. (a) (8 pts) Define the concept of spin crossover, using a generic octahedral complex (ML_6) for context.
   (b) (6 pts) What d-electron counts for the metal center are possible for observing spin crossover?
   (c) (6 pts) How would spin crossover be observed in a UV-vis spectroscopy measurements?
3. (a) (10 pts) Define the concept of linkage isomerism in an inorganic compound.
   (b) (10 pts) Give two examples of ligands that exhibit linkage isomerism, including a mechanism for switching the
4. (a) (5 pts) A small amount $D_3N(l)$ (D = deuterium) is added to NH_3 (l), and equilibrates to form H_2DN (aq). Sketch the structure of H_2DN (aq).
   (b) (5 pts) What is the point group of the H_2DN (aq) molecule?
   (c) (5 pts) How many vibrational degrees of freedom does H_2DN (aq) have?
   (d) (5 pts) Derive the character table for the point group you assigned to H_2DN.
5. Given the attached Pourbaix diagram for W, give an example of the compounds you may find in each of the *passivation*, *corrosion*, and *immunity* regions and explain your answer. Each must be correct for the region to receive credit for the problem.



6. (a) The reaction KCl (s) -> K^+ (aq) + Cl^- (aq) has a solubility product equilibrium constant of roughly $K_{sp} = 10^2$ while the analogous reaction for CuCl (s) has $K_{sp} = 10^{-7}$. Use the concept of hard-soft acid-base theory to explain this difference.
7. How can one control the average size and shape of electrodeposited nanocrystals of, for example, Pd for electrocatalysis? Give two examples (10 pts each) of how you can use chemical or electrochemical parameters to do so.
8. (a) (12 pts) Covalent organic frameworks (COFs) are crystalline porous polymers composed of a three-dimensional (3D) network of organic building blocks. Describe **one** experiments that would yield measurements of the *average* surface area per mass of a COF material.
   (b) (8 pts) Describe an application of COFs that would benefit from a high surface area per mass of material.
9. Researchers can use x-ray reflectometry (XRR) to study the structure of thin films. Describe how these measurements are different from traditional powder x-ray diffraction (PXRD) measurements and give two examples of structural characterization measurements that can be performed with XRR but not PXRD.
10. Propose a design for a chemical vapor deposition reactor used to make crystalline germanium. Include details about the specific compounds and reactions involved (no credit without these!) and any practical considerations (temperature, gas environment, etc.)